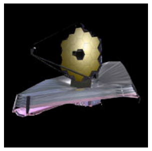

6010 - Spin doctor: unwinding stellar contamination from TRAPPIST-1

Cycle: 3, Proposal Category: AR

INVESTIGATORS

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OBSERVATIONS

ABSTRACT

In Cycles 1 and 2, a total of 265 hours were allocated for spectroscopy of a single target: TRAPPIST-1. This M8V star hosts seven transiting Earth-sized planets in a resonant chain, and its planets may be the first rocky exoplanets to have their atmospheres characterized. Transmission spectra collected with JWST can measure atmospheric compositions for the planets if and only if contamination from stellar atmospheres is absent or well-understood, but neither condition is met for TRAPPIST-1. Cycle 1 observations confirm sufficient time-variable stellar magnetic activity to prevent the detection of atmospheres for the outer planets. We will use archival JWST observations of TRAPPIST-1 to measure the properties of active regions, and the stellar contamination they produce, with two complementary approaches: (1) chromatic rotational modulation in the out-of-occultation spectra; and (2) transits of planets without thick atmospheres. We will determine which planetary spectral features are least affected by stellar contamination, and which climates or compositions can be detected/falsified with JWST observations given the ubiquitous stellar contamination.

OBSERVING DESCRIPTION